13	(a)	-4	B1	cao	
	(b)	$2r - 3t$	M1	for a^{2r} or a^{3t}	
			A1	$2r - 3t$ oe	Accept answer written as u^{2r-3t}
	(c)	$16\sqrt{2}$ or $8\sqrt{8}$	M1	for $8^{\frac{3}{2}}$ or 22.62... or $n^2 = 8^3$ or $n^{\frac{1}{3}} = 8^{\frac{1}{2}}$	
			A1	$16\sqrt{2}$ or $8\sqrt{8}$	Accept the answer written as $\sqrt{512}$

Documented scheme clarification

1MA1_MOCK_SET_04_2H_Q13

Original source

scheme page 9, SHA256
1b50aa06783c6cc171173cc3d7916bc8bad688a9b8e3c7e4c89ccac00764a3a8

Original wording

The extra guidance in(b) accepts a malformed power with base u.

Correction / interpretation

In (b), (a^r)^2 / (a^t)^3 = a^(2r-3t), so the requested expression is u=2r-3t. The extra guidance displaying u raised to 2r-3t is a notation error; the main answer is correct.

Reason

Index laws multiply powers then subtract exponents on division.

Sol authority

6556bd9484cf36b0c354ee080554a4708aede7b6e7124c68cbeb9eaa1e057175